

Special section

Shallow void, tunnel, and other anomaly detection — Introduction

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Voids may be natural or human made, have different shapes, and exist in various geological environments at depths of 1 m to much greater. Their only common property, which is a sharp contrast with their surroundings, has been the focus of decades of research aiming at their detection. Nonetheless, no definitive “silver bullet” has been found for this yet unresolved challenge, which is essential for engineering, security, and various academic disciplines counting on near-surface geophysical methods.

This special section reports the newest approaches attempting to solve this difficult problem. It contains 13 papers, beginning with a historical review of seismic methods used for void detection. A group of four papers focuses on imaging using diffracted waves, which are a prominent feature of the targeted voids. Three papers explore novel seismic signatures related to the existence of voids, and one uses full-waveform inversion to detect voids by their lower velocity. The last group, containing four papers, reports developments in electromagnetic and ground-penetrating radar (GPR) methods. More details about the published papers, organized by groups, can be found below.

Historical perspective

Sloan et al. provide a history of using active seismic methods to detect subterranean voids, from refraction and reflection methods available in the 1960s to more advanced techniques, such as full-waveform inversion, being applied today. The breadth of methods and techniques applied over the past 50 years demonstrates the advances in data acquisition, data processing, and computing that are enabling smaller and smaller anomalies to be detected.

Diffraction imaging

Liu et al. present an interferometric-based workflow for imaging shallow subsurface objects represented by weak diffractions hidden behind dominant surface waves (in their case, Love waves). The authors illustrate the robustness of their proposed workflow using both synthetic data and field data sets.

Abukrat and Reshef apply conventional shallow seismic methods on distributed acoustic sensing (DAS) data to conduct velocity analysis and successfully detect a subsurface void. In addition, the authors utilize the directional sensing of standard DAS arrays to address the problem of wave-mode separation.

Bakhtiari Rad and Hickey present a novel processing workflow for diffraction separation in the near surface based on wavefront attributes, which are determined using the common-reflection-surface method.

Rochlin et al. present an algorithm that can significantly improve the detection and characterization of linear elements of the subsurface using an edge diffractor operator. For each subsurface point under consideration, the authors’ algorithm accumulates diffraction energy from the entire linear diffractor, allowing accurate detection and characterization in sparse acquisition and low signal-to-noise scenarios.

Unique seismic signatures

Gritto et al. present new seismic imaging techniques that use seismic resonant waves to detect and locate an air-filled vertical water collection shaft; the approach is based on active-source seismic data and ambient-noise data. The techniques may be applied to other classes of near-surface strong inhomogeneities, such as vertical mine shafts, karst, or near-surface cavities.

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Peterie et al. investigate body-to-surface-wave conversion at a 3 m deep tunnel and develop two methods using converted surface waves to detect tunnels or other shallow anomalies. Converted surface-wave imaging in the time domain and surface-wave amplitude analysis in the frequency domain are used to confidently and independently locate the tunnel in real data.

Glaser et al. demonstrate an approach for the detection of voids below the practical depth of investigation of surface-based geophysical methods. The authors' work shows that secondary features associated with dewatering said voids can be used to locate the void itself.

Full-waveform inversion

Mecking et al. show a viscoelastic SH-full-waveform inversion applied on an SH-seismic data set to resolve a known but inaccessible tunnel system below an ancient burial mound. A new FWI workflow is presented, which focuses on the reflected signals by combining time- and offset-windowing approaches with a new AGC-objective function allowing the resolution and differentiation of air-filled voids and unstable zones of decompacted sediments.

Electromagnetic methods

Liu et al. develop a car-mounted 3D GPR system, as well as a data processing and imaging algorithm that aid the accurate

interpretation of GPR data recorded in complex urban environments, and have detected more than 100 road cavities in three Chinese cities during the past year. Field results indicate that 3D GPR can estimate the location, depth, lateral size, and inner height of the detected cavity with satisfactory accuracy, suggesting a potential tool to mitigate the disaster of road collapses in China and other countries.

Neese et al. study crosshole electromagnetic tunnel detection using a resonant coil system by evaluating analytic expressions for simplified models. When the best-suited (typically below GPR) frequencies are used, tunnel detection may be possible over a wider range of earth media than previously reported.

Dehiya examines the applicability of electromagnetic gradiometry in tunnel detection using a 3D forward modeling algorithm. Detailed numerical experiments are presented to evaluate the performance of different source-receiver configurations using various tunnel model scenarios.

Zou et al. propose a precision and accuracy method to inspect the interlayer debonding of airport pavement by using a multistatic GPR system. The proposed approach is easier to understand, and the measurement process is much simpler, allowing it to be implemented in real-time inspection.